Bertrand Competition with Reinforcement Learning: A Soft Demand Extension

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Traditionally, strategic interactions between rival firms in a duopoly have relied on classic economic models, such as those proposed by Cournot and Bertrand, which typically assume that players are perfectly rational agents with access to complete information. In these theoretical scenarios, firms converge toward a Nash Equilibrium, a state where no competitor has an incentive to change their strategy unilaterally. However, the rapid digitalization of global markets is changing these dynamics: today, many firms delegate strategic decision-making to automated algorithms that set prices in real-time based on high-frequency data (Cao & Hu, 2024).

This transition from human intuition to machine calculation raises an essential question: can independent algorithms, designed solely to maximize individual profits, learn to coordinate their actions without explicit communication? This phenomenon, known as "algorithmic collusion", challenges current antitrust enforcement. If algorithms independently reach prices above competitive levels, traditional legal definitions of collusion become ineffective, as there is no explicit agreement between human managers, leaving a significant gap in current regulations.

This paper explores these dynamics by applying advanced Reinforcement Learning (RL) techniques to the pricing duopoly problem. Following Calvano et al. (2020) approach, it's analyzed if independent algorithms can learn price strategies without human leading. The Deep Deterministic Policy Gradient (DDPG) algorithm is used, selected for its ability to operate effectively in continuous action spaces. Unlike discrete Q-learning, DDPG allows agents to select any price within a continuous range, coherently with the flexibility of real-world digital markets. By utilizing an Actor-Critic architecture, the agents can learn to approximate complex strategies and adapt to dynamic, stochastic market environments (Lillicrap et al., 2015).

The report is organized as follows:

- **The Basic Model**: a fundamental market characterized by homogeneous goods and a "winner-takes-all" demand structure was established. Then, to find the classic Bertrand Nash Equilibrium under varying levels of cost volatility, a manually implemented DDPG algorithm was tested.
- **The Extension**: a Soft Bertrand model was implemented using a Multinomial Logit demand structure. This increases economic realism by accounting for "soft" competition where consumers do not immediately switch for marginal price differences. This section also introduces Self-Play training methodologies.
- **Implementation and Analysis**: the software architecture, neural networks, and the training process are then detailed for the model's implementation. Finally, the analysis shows if agents converge toward the theoretical Nash Equilibrium or if there are signs of autonomous cooperation and price signaling.

The main goal of this study is to determine whether AI agents strictly follow traditional economic rules or if the introduction of demand smoothing and advanced training cycles fosters autonomous behaviors that deviate from standard competition.

Basic Model

Bertrand Competition

The basic model is built following the Bertrand Duopoly theory, where two firms compete by setting prices for a homogeneous product. Unlike the Cournot model, where firms compete on quantity, the Bertrand model assumes that consumers are price-sensitive and will always choose the cheapest option (Tirole, 1988).
In this study implementation, the market is defined by:

- **Homogeneous Goods**: there is no perceived difference between the products of Firm 1 and Firm 2.
- **Linear Demand**: the demand function is assumed to be linear.
- **"Winner-Takes-All" behavior**: the demand allocation follows a discontinuous logic, meaning that the cheaper firm captures every customer, while the more expensive one sells nothing; this forces both algorithms to cut prices until they reach the cost of production.

The One-Shot Game Nature

A crucial aspect of the basic model is that it is treated as a repeated One-Shot game.
In each round, the agents set their prices based only on the current state (the marginal cost) without a memory of past rounds affecting the immediate demand.
Then, to test the robustness of the agents, a stochastic marginal cost was introduced. In each round, it is sampled from a uniform distribution and this forces the agents to learn a policy rather than a fixed number, as they must adapt their price to the fluctuating cost.

Solution Concept: The Nash Equilibrium

The theoretical solution for this model is the Bertrand Paradox. The paradox lies in the fact that, even in a duopoly, the outcome is identical to that of perfect competition, where price equals marginal cost and firms earn zero economic profit. This occurs because the products are perfectly homogeneous; even a tiny price difference shifts the entire market demand to the lower-priced firm, creating a continuous downward pressure on prices.
Even with only two firms, the Nash Equilibrium occurs when both firms set their price equal to the marginal cost. At this point, economic profit is zero. Any price higher than the marginal cost would be undercut by the rival, and any price lower would result in a loss.

Extension

The second model is a smoothed extension of the classic Bertrand framework: instead of assigning the entire demand to the firm with the lowest price, which generates a fully discontinuous rule that is not amenable to continuous control methods, it introduces a logit allocation that distributes demand probabilistically based on price differences. Total demand remains linear:

$$D = \max(0, a - \min(p_1, p_2)) \quad (1)$$

but the market share is determined via a logistic function governed by τ, so that small price variations lead to gradual changes in quantities and profits. This smoothing removes discontinuities, provides useful gradients, and transforms the game into a stable and differentiable environment where algorithms like DDPG can reliably learn equilibrium strategies and self-play dynamics. Importantly, the use of logit demand is not only a

computational regularization but also well-grounded in economic theory. Morrow and Skerlos (2012) prove the existence of Bertrand–Nash equilibrium prices under logit demand in a very general setting, without relying on restrictive assumptions such as firm symmetry, single-product firms, or concavity of utility in prices. Their results show that smoothing demand through a logit structure resolves the discontinuities inherent in the classic Bertrand model, yielding well-behaved pricing equilibria.
In this model, τ governs the degree of price sensitivity in demand allocation between firms. Demand is split according to a logit rule, where s_1 represents firm 1's market share.

$$s_1 = \frac{1}{1 + \exp\left(\frac{p_1 - p_2}{\tau}\right)} \quad (2)$$

A small value of τ implies that consumers are highly price-sensitive, while a large value indicates lower sensitivity to price differences. As τ approaches zero, the model converges to the classic Bertrand outcome, in which the lowest-priced firm captures nearly the entire market. Conversely, as τ increases, demand becomes smoother, allowing firms to retain a positive share even when charging higher prices. In the current specification, τ = 0.2, placing the economy in a regime of strong but not extreme price competition. This setup closely approximates Bertrand competition while preserving continuous and differentiable payoffs a property crucial for DDPG-based learning. From an economic perspective, this smoothing can be interpreted as arising from search frictions, switching costs, or imperfect consumer information, which prevent consumers from responding perfectly to small price differences.

Model Implementation

The Environment

For the implementation of both model environments, the base model and its extension, the PettingZoo library was used.
The base environment models a Bertrand duopoly with homogeneous goods, meaning that consumers perceive the two products as perfectly identical and base their purchase decision solely on the prices. As a result, market demand is driven by price competition.
The environment is initialized with a single demand parameter a , which represents the maximum willingness to pay in the market and determines the size of potential demand. In particular, the parameter is set to $a = 40$, and if the market exceeds this value, the total demand drops to zero.

In each iteration, the two firms simultaneously choose their price, denoted by p_1 and p_2. The total market demand depends on the lowest price offered, $p_{min} = \min(p_1, p_2)$ and follows a linear demand function, defined in equation 1. This implies that the demand decreases linearly as the lower price increases and it becomes zero when the price reaches $a = 40$.

Demand is then allocated between firms according to standard Bertrand competition rules, allowing them to determine the quantity sold by each company:

- If $p_1 < p_2$, firm 1 captures the entire market demand, leaving zero demand to its competitor.
- If $p_1 > p_2$, firm 1 captures no demand and firm 2 serves the entire market.
- If $p_1 = p_2$, total demand is split equally between them.

Given the quantity sold q_i, each firm's profit is computed as:

$$\pi_i = (p_i - c)q_i \quad (3)$$

where c is the current marginal cost of production.

This environment is intentionally simplified, serving as a benchmark setting, allowing to isolate the pure effects of price competition in a homogeneous goods market, before introducing more complex demand.
Based on this basic Bertrand environment, we introduce an extension that replaces the standard "winner takes all" demand allocation with a smooth logit demand function, which ensures differentiable payoffs and makes learning with gradient-based reinforcement learning algorithms challenging.

The action space is continuous: each agent chooses a price p_i, constrained in the interval:

$$p_i \in [c, p_{max}]$$

where:

- $c = 2.0$ is the marginal cost of production, common to both firms.
- $p_{max} = 20.0$ is the maximum admissible price that consumers are willing to pay.

The observation space of firm 1 defines the information available to the agent when choosing its prices:

$$obs = (a, b, c, p_2)$$

where:

- $a = 10$ is the baseline demand intercept. It represents the maximum market demand when the price is zero.
- $a_{noise} = 1.0$, meaning that the market intercept can vary moderately around its mean in each reset. This stochasticity makes the learning problem more realistic and helps reinforcement learning agents adapt to varying demand conditions.
- $b = 1.0$ measure how fast the demand decreases as price increases.
- $c = 2.0$ is the marginal cost of production.
- p_2 is the opponent's price.

In the extended environment, total market demand remains linear, with respect to the lowest price, as defined in equation 1. However, instead of allocating demand discontinuously, the environment uses a smooth logit allocation. The share of demand for firm 1 is calculated as defined in equation 2, where τ is the temperature parameter that controls the smoothness of the allocation.
The quantity sold by each firm is then:

$$q_i = D_i * s_i \quad (4)$$

and the profit is as defined in equation 3.

The smooth allocation produces the following effects:

- If $p_1 < p_2$,: firm 1 is cheaper, $s_1 > 0.5$, meaning the firm 1 takes most of the market.
- If $p_1 = p_2$: firms split the market exactly 50/50 ($s_1 = s_1 = 0.5$).
- If $p_1 > p_2$: firm 1 is more expensive, so $s_1 < 0.5$, meaning the firm 1 takes less market than firm 2.

Compared to the original model, this new approach ensures that the payoffs vary smoothly with price, while still preserving the basic intuition of Bertrand competition: lowest prices attract more demand.

The Deep Deterministic Policy Gradient (DDPG) algorithm

The Deep Deterministic Policy Gradient (DDPG) algorithm is an actor-critic model-free, off-policy algorithm specifically designed for continuous action spaces. The agent consists of two neural networks:

- An **actor**, which proposes a price given the current state.
- A **critic**, which evaluates how profitable that price is by estimating the expected profit associated with a given state-action pair.

The actor network has a hidden layer with 64 neurons and uses a Sigmoid activation function in the output layer, ensuring the chosen action is bounded. Also, the critic network contains two hidden layers with 64 neurons and a linear output layer that produces a scalar value, representing the expected profit of the firm for a given state-action combination.

During interaction with the environment, the agent stores past experiences in a **replay buffer**:

$$(s_t, a_t, r_t)$$

where:

- s is the observed state,
- a is the chosen price,
- r is the realized profit.

The replay buffer allows the agent to sample experiences randomly during training, which:

- Breaks temporal correlations in the data.
- Improves learning stability.
- Enables off-policy learning.

At each training step, a minibatch of experiences is sampled from the replay buffer. The critic is updated first. For each sample transition, the critic computes the predicted Q-value.
The critic's loss is defined as the mean squared error between the predicted Q-values and the observed rewards:

$$L_c = (Q(s,a) - r)^2 \quad (5)$$

By minimizing this loss, the critic learns to approximate the firm's expected profit function given market conditions and pricing decisions. After updating the critic, the actor is updated using the deterministic policy gradient. The actor's objective is to choose prices that maximize the critic's estimated profit. This is achieved by minimizing the following loss function:

$$J(\theta) = E\big[Q\big(s, \mu(s)\big)\big] \quad (6)$$

The actor adjusts its pricing policy in the direction that increases the critic's predicted profit. The critic provides the gradient information necessary for improving the pricing strategy.

After the updates, the learning process is embedded in a market simulation in which two DDPG agents repeatedly interact in a Bertrand duopoly. In each round, the marginal cost is drawn as follows:

$$c_t = 10 + u_t, u_t \sim \mu(-1,1) \quad (7)$$

This introduces cost uncertainty, forcing agents to learn pricing strategies that are robust to fluctuations in production costs.

During training, agents follow an **epsilon-greedy exploration** strategy:

- With probability epsilon, firms choose random prices.
- With probability 1-epsilon, firms follow their learned pricing policy (actor network).

The exploration rate decreases linearly over time:

$$\varepsilon_t = \max\left(0{,}1 - \frac{t}{12000}\right) \quad (8)$$

This ensures sufficient exploration in early stages and converges to stable pricing strategies in later rounds, allowing the agent to gradually rely more on its learned policy.

Each realized interaction (s_t, p_t, π_t) is stored in the agent's replay buffer. After each round, both agents perform one training step:

- The critic is updated to better approximate the profit function.
- The actor is updated to improve the pricing policy.

This corresponds to **simultaneous learning by competing firms**, where each firm adapts the observed behavior of its rival.

Through training in the Bertrand duopoly environment, the agents learn that setting a lower price increases market share but reduces margins, while setting a higher price increases profit per unit but raises the risk of losing demand to the competitor. As a result, agents internalize the strategic interaction between price, demand and competition.
From an economic perspective, the **actor network** learns a pricing rule that maps observable market conditions into optimal pricing decisions. At the same time, the **critic network** learns to approximate the firm's profit function under competitive interaction by evaluating the expected profitability of different price choices. The continuous interaction between the two allows the agent to converge toward the best response pricing policy.

Analysis

In this section are analysed the performance of the DDPG agents within the Basic and the Extended environment. The primary focus in the Basic Bertrand model is to observe how the agents successfully converge toward the theoretical Nash Equilibrium, adapting their pricing strategies to different levels of market volatility.

Convergence Toward the Nash Equilibrium

The simulation results indicate that the DDPG agents successfully learn the underlying economic rule of the Bertrand competition. After an initial phase of exploration (characterized by high epsilon values and random pricing), the agents progressively lower their prices as they realize that any price significantly above the marginal cost results in zero demand due to the "winner-takes-all" structure.
As observed in the simulation runs, the agents converge toward a state where the price of the agent 1 is approximately equal to the price of the agent 2 and they are approximately equal to the marginal cost. This confirms that the neural networks have effectively approximated the Bertrand Paradox: even in a duopoly, the competitive pressure forces prices down to the marginal cost, resulting in near-zero unit margins.

Impact Of Cost Volatility: High vs Low Variance

A key observation in this analysis is the difference in training stability between high and low cost-variance scenarios:

- **Low Variance Scenario (marginal cost +- 1):** When the marginal cost is stable, the learning signal is much clearer for the Critic network. The results show a cleaner convergence where the unit margin stabilizes very close to zero with minimal oscillations. The agents quickly learn to track the marginal cost with high precision.
- **High Variance Scenario (marginal cost +- 10):** With high volatility, the agents face a more complex environment. While they still learn to follow the general trend of the marginal cost, the price evolution shows significant noise. The agents occasionally overcome or undercome the equilibrium price, as the rapidly changing state makes it harder for the neural network to generalize the optimal policy.

Figure 1 shows the low variance scenario, clearly identifiable because the marginal cost stays consistently close to 10, that is the value of the marginal cost.

Figure 1. Price Evolution vs Marginal Cost

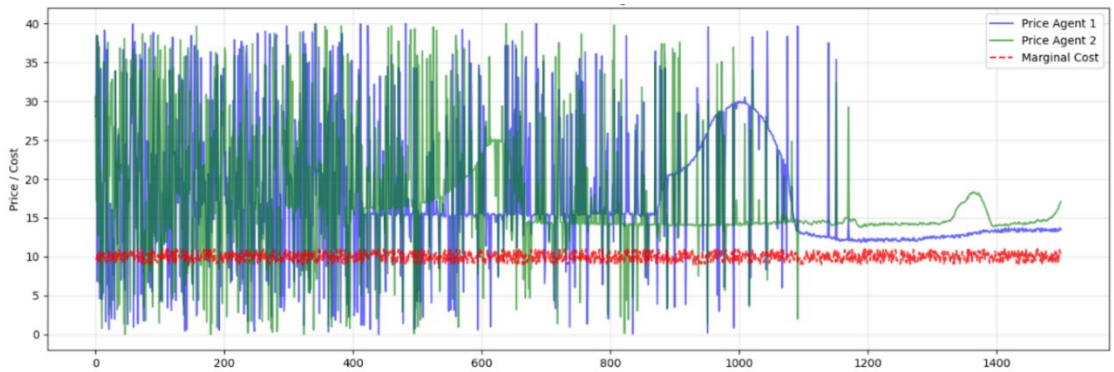
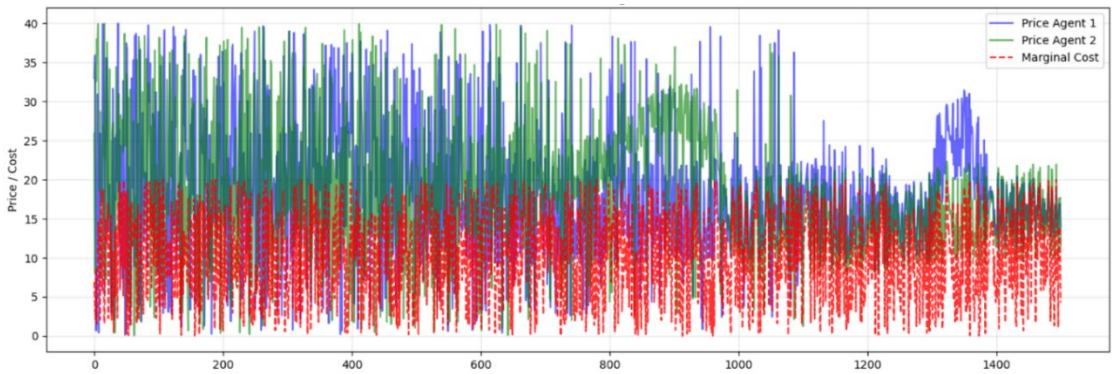


Figure 2 instead refers to the high variance scenario, where the marginal cost rapidly changes between 0 and 20.

Figure 2. Price Evolution vs Marginal Cost

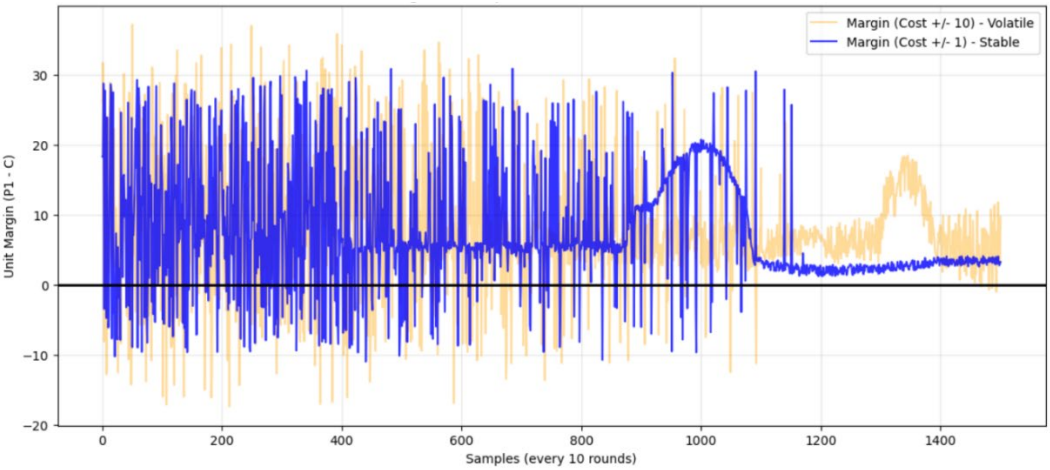


Unit Margin and Profit Distribution

The analysis of the unit margins (price - marginal cost) reveals that during the early stages of training, margins are highly volatile and frequently negative, as agents underprice the cost to capture the market. However, as training progresses and the exploration rate decays, the margins stabilize in a narrow positive range just above zero.

This suggests that, while the agents reach a competitive equilibrium, they tend to maintain a very small "buffer" above the marginal cost. This behavior can be attributed to the stochastic nature of the DDPG algorithm and the inherent difficulty of reaching a perfect mathematical zero in a continuous action space.
The graph below shows the comparison between the volatile market, represented in blue, and the stable one, represented in orange.

Figure 3. Convergence Comparison: Volatile vs Stable Market



Model Extension

Figures 4 and 5 show the evolution of prices and profits across self-play rounds. Initial rounds exhibit high volatility as agents explore the pricing space and respond to poorly anticipated rival behavior. Over time, prices converge symmetrically to a stable level strictly above marginal cost. This outcome reflects the presence of demand smoothing induced by the logit specification, which softens price competition and yields an interior Bertrand–Nash equilibrium. Correspondingly, profits stabilize at positive and symmetric levels, indicating that firms learn to avoid aggressive undercutting while still competing for market share.

Figure 4. Price Dynamics Across Self-Play Rounds

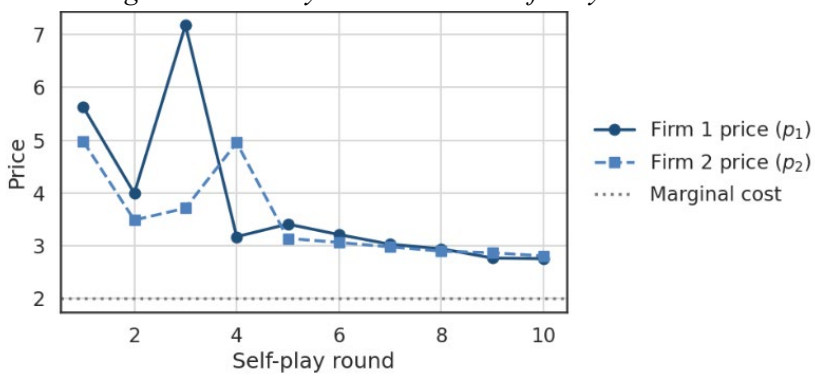
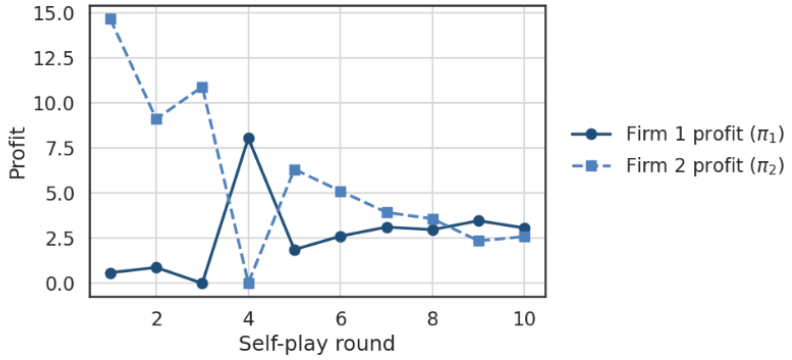


Figure 5. Price Dynamics Across Self-Play Rounds



Although the model approximates Bertrand competition, prices do not converge exactly to marginal cost and profits are not identically zero. This outcome is driven by several features of the learning environment. First, the use of a logit demand structure with $\tau > 0$ implies that firms retain a strictly positive share of demand even when charging slightly higher prices than their rival. As a result, marginal undercutting incentives are softened relative to the knife-edge Bertrand case.

Second, firms are represented by learning agents that rely on stochastic exploration through action noise, a standard feature of continuous-control reinforcement learning algorithms. This exploration prevents exact convergence to a single deterministic price and generates small but persistent fluctuations around the competitive benchmark.

Third, self-play learning creates a non-stationary strategic environment in which each agent continuously adapts to the evolving policy of its opponent. Temporary mismatches in learned strategies can therefore give rise to short-lived positive profits, even in a highly competitive regime.

Taken together, these factors imply that prices and profits fluctuate within a narrow neighborhood of the Bertrand competitive benchmark rather than converging exactly to it. Importantly, these deviations do not reflect stable collusive behavior, but instead arise from demand smoothing, intrinsic exploration, and ongoing strategic adaptation.

Conclusion

To conclude, the study shows that independent Deep Deterministic Policy Gradient (DDPG) agents can successfully internalize the fundamental principles of Bertrand competition.
In the basic model, DDPG agents consistently converge toward the Nash Equilibrium, setting prices close to marginal cost, and earning near-zero profits, reproducing the Bertrand Paradox.
When the model is extended to include a smooth logit demand structure, the learning process becomes more stable and economically realistic. By eliminating the “all or nothing” discontinuity given by the classical demand function, the extension, allows the actor network to more effectively map market conditions to optimal pricing decisions, while critic network achieves a more precise approximation of the expected profit function.

Overall, the results suggest that Reinforcement Learning algorithm are not only capable of replicating established theoretical equilibria, but can also discover robust pricing strategies in dynamics and uncertain environment.

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